Bank Runs, Deposit Insurance, and Liquidity

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PUBLICATION YEAR	1983
JOURNAL	Journal of Political Economy, vol. 91, no. 3 (June 1983), pp. 401–419
DOI	10.1086/261155
URL	journals.uchicago.edu/doi/10.1086/261155
RESEARCH AREA	Banking · Financial Intermediation · Theory of Financial Crises
SERIES	Banking
PAPER TYPE	Theoretical — general-equilibrium contract and mechanism-design model

ABSTRACT

This review examines the first fully micro-founded model of why banks issue demand deposits and why such deposits leave banks vulnerable to runs. Consumers face a privately observed liquidity shock that ordinary contingent-claims markets cannot insure, because the shock is not publicly verifiable. A bank pooling depositors' claims on a partially illiquid technology and promising a fixed early-withdrawal value can replicate the allocation a planner who observed types directly would choose. The same contract, however, admits a second, self-fulfilling equilibrium in which every depositor withdraws immediately, because the sequential-service constraint makes the bank's assets worth more to early claimants than late ones once a run begins. Suspension of convertibility restores the efficient equilibrium only when withdrawal demand is deterministic; no bank contract obeying sequential service can achieve full efficiency once that demand is stochastic; and government deposit insurance financed by ex post taxation can implement the unconstrained optimum uniquely, because taxation is not bound by the queue. The result reframes runs as an equilibrium phenomenon inherent to liquidity transformation, not a symptom of insolvency.

1. Introduction

Diamond and Dybvig (1983) ask a question that had not, before this paper, been given a rigorous equilibrium answer: why do banks fund themselves with demand deposits — liabilities that can be withdrawn at par, on demand, in the order requested — when doing so leaves them exposed to the possibility that all depositors withdraw at once and force the bank into a fire sale of illiquid assets? The empirical fact of bank runs was well documented (Friedman and Schwartz 1963 remains the authoritative account of the 1930s panics), but the existing theory of financial contracting offered no reason why banks should write contracts with this particular fragility built in, as opposed to some other, safer arrangement.

The question matters for two distinct reasons, one positive and one normative. Positively, any theory of banking has to explain why intermediaries hold illiquid assets (loans, long-term investments) against liquid liabilities (deposits) rather than simply matching the maturity of assets and liabilities, which would be trivially safe. Normatively, if runs are a structural feature of the optimal contract rather than a market failure to be regulated away, then the policy question changes from "how do we stop banks from making unstable promises" to "how do we preserve the liquidity-transformation service while eliminating its bad equilibrium." The paper was written in 1983, against the backdrop of U.S. banking deregulation and the early stages of the savings-and-loan crisis, and the authors are explicit that they see the question as a live policy matter, not only a theoretical curiosity.

Before this paper, the literature offered three separate pieces, none of which fit together into a single model. Historical and monetary accounts (Friedman and Schwartz 1963; Fisher 1911) described runs and their real economic costs but did not derive them from optimizing behavior. Deposit-insurance-pricing models (Merton 1977, 1978; Kareken and Wallace 1978) treated the probability of bank failure as exogenous and asked only how to price insurance against it, which presumes rather than explains any welfare role for insurance. And general risk-sharing theory (Arrow-Debreu contingent claims, as characterized through the lens of Patinkin 1965, Tobin 1965, and Niehans 1978 on liquidity) had no natural way to handle a risk that is real to the individual but privately observed and therefore not directly contractible. Bryant (1980) came closest, sketching a reduced-form model of reserves, runs, and deposit insurance, but without a fully specified mechanism-design argument for why the demand-deposit contract itself is the right response to the underlying informational friction.

Diamond and Dybvig close this gap by building an economy in three periods with a single good, in which each of a continuum of ex ante identical agents privately learns, at the interim date, whether she is an early consumer ("type 1") or a late consumer ("type 2"). Because this realization is private information and cannot be verified, no state-contingent insurance contract conditioned directly on type can be written; agents can only be offered a menu that is incentive compatible given what they alone know about themselves. The productive technology available to the economy is illiquid: it pays a high return if held to maturity but only returns principal if liquidated early. The authors show, first, that a bank contract promising a fixed sum to anyone who withdraws early can implement the same risk-sharing allocation a planner with full information would choose, and that this is a genuine equilibrium of the contract. They show, second, that the identical contract has another equilibrium — a bank run — in which everyone withdraws immediately, a self-fulfilling outcome that is worse for every agent than not banking at all. They then examine what can be done about the coexistence of these two equilibria: suspension of convertibility removes the bad equilibrium when the bank knows in advance exactly how many depositors will want to withdraw early, but this device fails once that number is itself uncertain (Proposition 1). Government deposit insurance funded by an after-the-fact tax succeeds where suspension fails, because a tax authority, unlike a bank subject to sequential service, can condition payments on the total realized withdrawal rate rather than only on an individual's place in the queue (Proposition 2).

The result matters because it converts "banks are fragile" from an empirical regularity into a theorem: fragility is not evidence that banks are mismanaged or overleveraged, but a structural implication of providing liquidity insurance at all under a first-come-first-served payment rule. That reframing set the research agenda — on deposit insurance design, lender-of-last-resort policy, and equilibrium selection in coordination games — that the banking-theory literature has worked through for the four decades since.

2. Research Problem and Hypothesis

2.1 Research Question

Formally: can a contract between a bank and a continuum of ex ante identical depositors, who face privately observed and non-verifiable heterogeneity in the timing of their consumption needs, implement the full-information-optimal sharing of the return on a partially illiquid technology as an equilibrium outcome — and, if so, is that implementation the *unique* equilibrium of the contract, or does it necessarily coexist with an inferior, self-fulfilling equilibrium in which every depositor withdraws immediately? A subsidiary question follows directly: what institutional devices (suspension of convertibility, government deposit insurance) can eliminate the inferior equilibrium, and under what conditions on the environment (in particular, whether aggregate withdrawal demand is deterministic or stochastic) do those devices succeed or fail?

2.2 Motivation

The question is important on both theoretical and applied grounds. Theoretically, it supplies banks with an explicit economic *raison d'etre* — the transformation of illiquid assets into liquid liabilities under private information about liquidity needs — that is distinct from, and complementary to, the delegated-monitoring rationale for intermediation that Diamond develops separately (working paper 1980, published 1984). Applied, the model gives the first formal account of runs as a coordination failure among rational agents rather than as either irrational panic or a symptom of insolvent fundamentals, which has direct implications for how deposit insurance, suspension rules, and lender-of-last-resort facilities should be designed. Every subsequent formal treatment of bank fragility, from the global-games refinements of the 2000s to the analysis of the March 2023 U.S. regional-bank runs, takes this paper's multiple-equilibrium framing as its starting point or its foil.

2.3 Existing Literature

Three strands existed prior to the paper, none of which the authors regard as sufficient on its own. The historical-empirical strand (Friedman and Schwartz 1963) established that runs did real damage during the Great Depression and that policy (the founding of the Federal Reserve, the later introduction of federal deposit insurance) was shaped by that experience, but offered description rather than a testable equilibrium theory. A second, older strand associated with Fisher (1911) attributes runs to a decline in the true value of a bank's assets below its fixed nominal liabilities — runs as a rational response to insolvency risk, not a coordination failure — a view the authors explicitly contrast their own with in the introduction. The deposit-insurance-pricing strand (Merton 1977, 1978; Kareken and Wallace 1978; Dothan and Williams 1980) modeled deposit insurance as a contingent claim to be priced given an exogenous default probability, which is a valuable actuarial exercise but is silent on why insurance should exist as a welfare-improving institution in the first place. Bryant (1980) is the closest predecessor, presenting a model of bank reserves, withdrawal risk, and deposit insurance with some of the same flavor, but without the formal, incentive-compatible mechanism-design derivation of the risk-sharing contract that is this paper's central technical contribution.

2.4 Research Gap

What was missing was a model that derives, from common micro-foundations, both (i) a positive economic role for banks issuing demand deposits, grounded in an explicit and realistic informational friction (privately observed liquidity type), and (ii) an equilibrium-theoretic account of why the very same contract that performs this useful function is also vulnerable to a self-fulfilling run, together with a precise statement of which policy interventions restore uniqueness of the good equilibrium and under what conditions.

2.5 Contribution

The authors state three contributions: that demand deposits can improve on competitive-market risk sharing; that the contract achieving this improvement has an undesirable run equilibrium; and that runs cause real economic damage rather than merely reflecting other problems.

> **CRITIQUE**
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> The first two claims are rigorously established within the model and constitute its lasting contribution. The third is comparatively under-demonstrated on its own terms: in the model, a run is costly for a purely mechanical reason — it forces liquidation of the entire illiquid technology at the inferior early return, interrupting production that would otherwise have continued to period two. This is a direct implication of the assumed payoff structure of the technology rather than an independent result requiring separate proof, and the model contains no channel (other banks, an interbank market, aggregate output outside the banking sector)

> through which this single-bank cost could propagate to "the economy" in the systemic sense the introduction invokes when citing the Great Depression. The claim that runs cause real damage is therefore better read as an interpretive gloss on a mechanical feature of the model's technology than as a freestanding finding on the same evidentiary footing as Propositions 1 and 2.

3. Theoretical Framework and Mechanism

3.1 Environment and Technology

Time runs over three dates, $T = 0, 1, 2$, and there is a single homogeneous good. A continuum of agents, identical as of $T = 0$, is each endowed with one unit of the good at $T = 0$ and nothing thereafter. A constant-returns productive technology converts input at $T = 0$ into output according to

$$1 \text{ unit at } T = 0 \rightarrow R \text{ units at } T = 2 \text{ if held to maturity, or } 1 \text{ unit at } T = 1 \text{ if liquidated early}$$

with $R > 1$. The choice of whether to liquidate is made at $T = 1$. This is the paper's single source of illiquidity: capital committed to the technology is not merely risky but structurally costly to unwind before maturity, since early liquidation forfeits the entire return premium $R - 1$. Diamond and Dybvig note that the qualitative results are unchanged if illiquidity instead arises from transaction costs of selling a long-lived asset early; the specific mechanical form is a modeling convenience, not the substantive assumption.

3.2 Preferences and the Private-Information Friction

At $T = 1$ each agent privately learns her type θ: with probability t she is type 1 (an early consumer, valuing consumption only at $T = 1$); with probability $1 - t$ she is type 2 (a late consumer, valuing consumption only at $T = 2$, or the sum of what she withdraws at $T = 1$ and stores costlessly plus what she withdraws at $T = 2$). Utility is

$$U(c_1, c_2; \theta) = u(c_1) \text{ if type 1}, \quad = \rho \cdot u(c_1 + c_2) \text{ if type 2}$$

where u is twice continuously differentiable, strictly increasing, strictly concave, satisfies the Inada conditions, and — critically — has relative risk aversion coefficient $-c \cdot u''(c)/u'(c) > 1$ everywhere. The discount factor satisfies $R^{-1} < \rho \leq 1$, so late consumption is discounted ($\rho \leq 1$) but not so heavily that the technology's return fails to compensate for waiting ($\rho R > 1$). Type is never publicly observable or verifiable, so no contract can be written directly on θ; any allocation must be self-selecting.

INFERENCE — this is the paper's central modeling choice, and it is what separates it from ordinary Arrow-Debreu insurance: the risk being insured (whether one turns out to be an early or late consumer) is idiosyncratic, privately observed, and realized after contracting, which is precisely the configuration under which competitive spot and forward markets alone cannot deliver full insurance, because no one but the agent herself ever knows which realization occurred.

3.3 The Competitive Benchmark (No Bank)

Absent any bank, constant returns to scale pin down competitive prices: the $T = 0$ price of $T = 1$ consumption is 1, and the $T = 0$ and $T = 1$ prices of $T = 2$ consumption are R^{-1}. Because there is no public signal on which to condition trade, the competitive equilibrium simply has every agent producing for her own consumption: a type 1 agent liquidates at $T = 1$ and gets $c_1 = 1$; a type 2 agent holds to maturity and gets $c_2 = R$. There is no insurance and no trade in equilibrium — each agent bears her own type-realization risk in full.

3.4 Full-Information Optimal Risk Sharing

Suppose, counterfactually, that a planner could observe each agent's type at $T = 1$. The planner's problem is to choose c_1^* and c_2^* to maximize ex ante expected utility subject to the economy's resource constraint. The solution satisfies

(1a) $c_2^* > c_1^*$ — late consumers, who can wait, are allocated more than early consumers under the optimum;

(1b) $u'(c_1^*) = \rho R \cdot u'(c_2^*)$ — the standard intertemporal first-order condition equating marginal utility of early consumption to the discounted, return-weighted marginal utility of late consumption;

(1c) $t \cdot c_1^* + (1 - t) \cdot c_2^*/R = 1$ — the period-0 resource constraint: one unit of endowment must fund the promised consumption of both types, where financing c_2^* at $T = 2$ costs only c_2^*/R in $T = 0$ resources because the technology multiplies invested principal by R.

Because $\rho R > 1$ and relative risk aversion exceeds unity everywhere, the paper proves (its footnote 3, by bounding $\rho R \cdot u'(R)$ below $u'(1)$ and using that u' is monotonically decreasing) that the optimum satisfies $c_1^* > 1$ and $c_2^* < R$: optimal risk sharing narrows the gap between what an early and a late consumer receive, relative to the no-insurance competitive outcome of $(1, R)$. This is the formal sense in which pooling has value: agents are willing to give up some expected return (moving c_2 below R) in exchange for insurance against the risk of being an early consumer (moving c_1 above 1).

ILLUSTRATIVE CALCULATION (REVIEWER'S OWN, NOT IN THE ORIGINAL PAPER)

To make the abstract comparative static concrete, consider a CRRA specification $u(c) = c^{1-\gamma}/(1-\gamma)$ with relative risk aversion γ, and the calibration $R = 1.20$, $\rho = 0.90$, $t = 0.25$, $\gamma = 3$ (chosen only for illustration; the paper reports no calibrated example of its own). Solving (1b)–(1c) numerically gives $c_1^* \approx 1.122$ and $c_2^* \approx 1.151$, against the autarky outcome of $(1.000, 1.200)$ — see Table 1 and Figure 2 below.

3.5 The Demand Deposit Contract and the Sequential-Service Constraint

The optimum in (1a)–(1c) requires knowing each agent's type, which is private information. The paper's central positive result is that a simple demand-deposit contract can nonetheless implement it as an equilibrium, without ever conditioning payment directly on type. Under the contract, an agent withdrawing at $T = 1$ receives a fixed sum r_1 per unit deposited, served in the order requests arrive, until the bank's assets are exhausted — the *sequential service constraint*. Formally, let f_j denote agent j's position in the withdrawal queue as a fraction of total deposits, and f the total fraction of deposits withdrawn at $T = 1$. Payoffs are

$$V_1(f_j, r_1) = r_1 \text{ if } f_j \le 1/r_1, \quad 0 \text{ otherwise} \tag{2}$$

$$V_2(f, r_1) = \max\{ R(1 - r_1 f)/(1 - f),\ 0 \} \tag{3}$$

V_1 is the payoff to a unit withdrawn at $T = 1$: the bank pays r_1 until its liquidated assets run out, at which point later claimants in the same withdrawal wave get nothing. V_2 is the pro-rata liquidation value, at $T = 2$, of what is left after paying out r_1 per unit to the fraction f who withdrew early: the bank has $(1 - r_1 f)$ units of principal remaining, invested and multiplied by R, divided among the remaining $(1 - f)$ depositors.

3.6 Equilibrium Multiplicity

Setting $r_1 = c_1^*$ reproduces the full-information optimum as one equilibrium: if type 1 agents withdraw at $T = 1$ and type 2 agents wait, then $f = t$, and substituting into (2)–(3) recovers $V_1 = c_1^*$ and $V_2 = c_2^*$ exactly — the "good" equilibrium. But for any $r_1 > 1$ a second, self-fulfilling equilibrium also exists: if every agent expects every other agent to withdraw at $T = 1$, then withdrawing early is each agent's best response too, because a late claimant would receive a share of a bank that has already liquidated everything at the inferior rate — a run. Figure 1 below plots V_1 and V_2 against f for the calibration above and shows the crossing point beyond which waiting no longer pays. The two equilibria are Pareto-rankable: the run equilibrium delivers every agent a risky payoff with mean 1, strictly worse than simply holding the asset directly, which is why banking under this contract is a positive-sum activity in the good equilibrium and a negative-sum one in the bad equilibrium. Setting $r_1 = 1$ eliminates the run equilibrium entirely ($V_1 \le V_2$ for all f) but also eliminates any liquidity-insurance benefit, since the contract then simply mimics direct asset ownership. The vulnerability to runs and the provision of liquidity insurance are, in this model, the same coin viewed from two sides: any $r_1 > 1$ that improves on autarky necessarily admits a run equilibrium.

3.7 Suspension of Convertibility

If the bank commits to serve no more withdrawals once a fraction $\hat{f}$ of deposits has been paid out at $T = 1$, and if $\hat{f}$ is chosen in the interval $[t, (R - r_1)/(r_1(R - 1))]$ with $r_1 = c_1^*$, then no type 2 agent ever wants to withdraw early regardless of what she expects others to do — waiting strictly dominates withdrawing for every possible realization of $f \le \hat{f}$. This makes the good outcome a dominant-strategy equilibrium, the strongest form of equilibrium stability the paper considers, and it is historically grounded: suspension of payments was a standard defense used by pre-deposit-insurance banks, and Friedman and Schwartz (1963) is cited as documenting that the Federal Reserve's restriction of banks' ability to suspend in the early 1930s likely made the ensuing panic worse rather than better.

3.8 Stochastic Withdrawal Demand and Proposition 1

Suspension works cleanly only because the bank, in that analysis, knows t in advance. The paper then lets the fraction of early consumers be a random variable $\tilde{t}$, unobserved by the bank and by any individual depositor before withdrawal decisions are made. The result is an impossibility theorem:

> **PAPER — PROPOSITION 1**
>
> No bank contract obeying the sequential-service constraint — that is, no contract in which payment to a withdrawer can depend only on her place in the queue and on realized aggregate withdrawals up to that point, not on private information others hold — can achieve the full-information-optimal risk sharing of (1a)–(1c) when $\tilde{t}$ has a nondegenerate distribution.

The proof is by contradiction in two steps. If the payment to an $T = 1$ withdrawer varied with the realization of $\tilde{t}$, then two type-1 agents facing different realizations would receive different consumption purely as a function of a variable they do not control and cannot observe, which cannot be part of an ex ante optimum in which all type-1 agents should be treated alike. If instead the $T = 1$ payment is held constant across realizations, the resource constraint (1c) forces the $T = 2$ payment to vary with $\tilde{t}$ in a way that is inconsistent with the first-order condition (1b), which pins down a unique relationship between $c_1^*(t)$ and $c_2^*(t)$ for each realization t. Either way, sequential service is structurally incompatible with full efficiency once aggregate withdrawal demand is uncertain.

3.9 Government Deposit Insurance and Proposition 2

The government, unlike the bank, can levy a tax after observing the realized withdrawal fraction f, including on agents who have already withdrawn — an instrument not available under sequential service. Consider a proportional tax schedule $\tau(f)$ on wealth withdrawn at $T = 1$, calibrated so that after-tax proceeds exactly replicate the full-information optimum for every realization of t:

$$\tilde{V}_1(f) = c_1^*(f) \text{ for } f \leq \bar{t}, \quad 1 \text{ for } f > \bar{t}$$

$$\tilde{V}_2(f) = R\{1 - [c_1^*(f) \cdot f]\}/(1-f) \text{ for } f \leq \bar{t}$$

where $\bar{t}$ is the largest possible realization of $\tilde{t}$, and excess tax revenue collected beyond what is needed to fund $T = 1$ withdrawals is reinvested in the bank to minimize forced liquidation. Because the government's tax can be conditioned on the full realization of f rather than only on an individual's queue position, it can be structured so that $\tilde{V}_1(f) \leq \tilde{V}_2(f)$ for every possible f:

> **PAPER — PROPOSITION 2**
> Demand deposit contracts combined with government deposit insurance financed by an optimally chosen tax achieve the unconstrained full-information optimum as the unique — in fact, dominant-strategy — Nash equilibrium.

CRITIQUE The economic content of Proposition 2 is not that deposit insurance provides a subsidy; a correctly priced actuarial insurance contract, of the kind Merton (1977, 1978) study, would not by itself change the equilibrium-selection problem. What does the work is that the taxing authority is not constrained by sequential service the way a bank is: it can act on aggregate information (f) that only becomes available after the fact, and it can retroactively adjust an individual's net payment even after she has withdrawn. This is a genuinely novel identification of *why* a public backstop can dominate any privately feasible contract — not because the government has deeper pockets, but because it has an informational and enforcement capacity (ex post, economy-wide, unconditional taxation) that a sequentially-serving bank structurally lacks. The result is conditional, however, on the tax being costless and unconstrained, a point the authors themselves flag as a caveat in the text (see §6 and §7 below).

4. Model Implementation: Contracts, Equilibria, and Extensions

NOTE ON STRUCTURE This paper contains no data, sample, or estimation strategy: it is a pure theory paper, and CLAUDE.md's default template section for "Data and Empirical Strategy" is adapted accordingly, following the instruction to emphasize theory and derivation for theoretical papers and not force irrelevant sections. In place of a data section, this section asks the analogous methodological question for a theory paper: where does the paper's identifying *logical* variation come from — that is, exactly which assumption is doing the work at each step of the argument, and what happens if that assumption is relaxed.

4.1 What Identifies the Good Equilibrium

The efficient equilibrium is identified by three jointly necessary features of the environment: (i) relative risk aversion strictly greater than one, which the proof in footnote 3 uses to sign $c_1^* > 1$ and $c_2^* < R$; (ii) $\rho R > 1$, which ensures type 2 agents genuinely prefer to wait under full information; and (iii) the self-selection (incentive-compatibility) property verified in the paper's footnote 4, that neither type envies the other's consumption bundle under the optimal contract, which is what allows a mechanism that does not observe type directly to still implement the type-contingent optimum.

4.2 What Identifies the Bad Equilibrium

The run equilibrium is identified by exactly one structural feature: the sequential-service constraint combined with $r_1 > 1$. Sequential service is what makes an individual's payoff depend on the beliefs and actions of others rather than only on the realized state of the world — it is the coordination device through which self-fulfilling expectations operate. Remove sequential service (allow the bank to pay everyone a state-contingent amount based on the full realization of withdrawals, as the government effectively does in §3.9) and the multiplicity disappears along with the constraint that generated it.

4.3 What Proposition 1 Isolates

Proposition 1's impossibility result isolates stochastic aggregate withdrawal demand, $\tilde{t}$, as the feature that defeats suspension of convertibility specifically — not the private-information friction on individual type, which is present throughout the paper and is not what breaks efficiency here. This distinction matters for interpreting the result: the paper is not saying private information about type is incompatible with efficiency (§3.5 shows it is not); it is saying that private information about type *combined with* public uncertainty about the aggregate distribution of types defeats any contract bound by sequential service.

4.4 What Proposition 2 Isolates

Proposition 2 isolates the ability to condition on ex post, economy-wide information as the feature that restores efficiency. The result is silent on financing feasibility in a world where taxation is distortionary, has an incidence unrelated to who caused the shortfall, or is politically constrained; the authors describe their own result as possibly "too strong" for exactly this reason. This is a genuine boundary condition on the theorem's applicability, not a minor caveat, because the entire empirical case for public deposit insurance schemes (flat-rate ex ante premia, not state-contingent ex post levies) departs from the financing instrument the proof actually uses.

5. Results: Formal Propositions and Illustrative Analysis

5.1 Summary of Formal Results

The paper's results are best read as a chain of four claims of decreasing generality. First, a demand-deposit contract with sequential service can implement the full-information optimum as one equilibrium when t is known (an existence result). Second, the same contract admits a strictly worse equilibrium for any $r_1 > 1$ (a non-uniqueness result). Third, suspension of convertibility restores uniqueness — as a dominant-strategy equilibrium, no less — conditional on t being known ex ante (a conditional fix). Fourth, once t is stochastic, no contract bound by sequential service can do what suspension did (Proposition 1), but a government with unconstrained ex post taxing power can (Proposition 2). None of these are statistical results with standard errors or sample sizes; they are proofs, and their "significance" is logical validity conditional on the stated assumptions, not statistical significance in the empirical sense.

5.2 Table and Figure Analysis

The original article contains no tables or figures — typical of Journal of Political Economy theory articles of this period, which relied on algebra and verbal proof rather than visual exposition. The table and figure below are the reviewer's own constructions, built strictly from the paper's equations (1)–(3) under the illustrative calibration introduced in §3.4, and are included because CLAUDE.md requires visual analysis where it improves understanding of a paper's mechanism; they are not reproductions of anything in the original text.

Table 1. Illustrative calibration of full-information optimal risk sharing versus autarky

Quantity	Autarky (no bank)	Full-information optimum
Early consumption, type 1 (c_1)	1.000	1.122
Late consumption, type 2 (c_2)	1.200 (= R)	1.151
Implied liquidity-insurance premium ($c_1^* - 1$)	—	+12.2%
Implied return sacrifice ($R - c_2^*$)	—	0.049

Source: reviewer's calculation, CRRA utility with $\gamma = 3$, $R = 1.20$, $\rho = 0.90$, $t = 0.25$, solving equations (1b)–(1c) above. Illustrative only; the calibration is chosen for exposition and is not fit to any observed banking data.

The table shows what §3.4 states algebraically: under this calibration, optimal risk sharing raises the consumption available to an early consumer by roughly 12 percent relative to holding the asset directly, financed by a reduction in the late consumer's payoff of about 4 cents per unit deposited, or roughly 4 percent of R. The asymmetry — a large proportional gain for the early state financed by a small proportional loss in the late state — is exactly what strictly concave utility with relative risk aversion above one implies: the marginal utility of consumption is far higher in the low-consumption (early) state, so a small transfer away from the high-consumption (late) state buys a much larger proportional improvement in the low state. This is a coefficient interpretation strictly internal to the model's own algebra, not an empirical magnitude; it does not by itself say anything about the actual size of liquidity-insurance benefits in real banking systems, which would require calibrating R, t, and risk aversion to observed data the paper does not use.

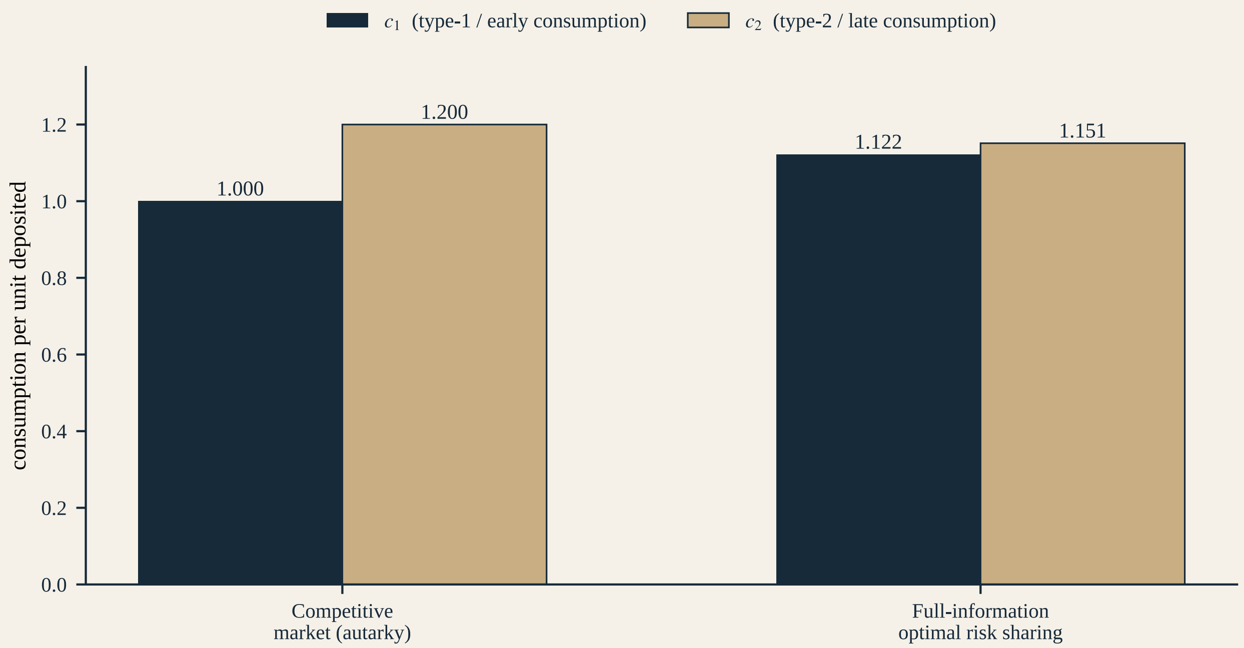

Figure 1. Autarky versus full-information optimal consumption, by type

Source: reviewer's calculation from equations (1a)–(1c), same calibration as Table 1. Not contained in the original paper. The figure shows a cross-sectional comparison of two counterfactual allocations for the same economy, not a time series or an estimated relationship; there is no sampling variation or confidence interval to report because the underlying object is a closed-form solution to a planner's problem, not an estimate.

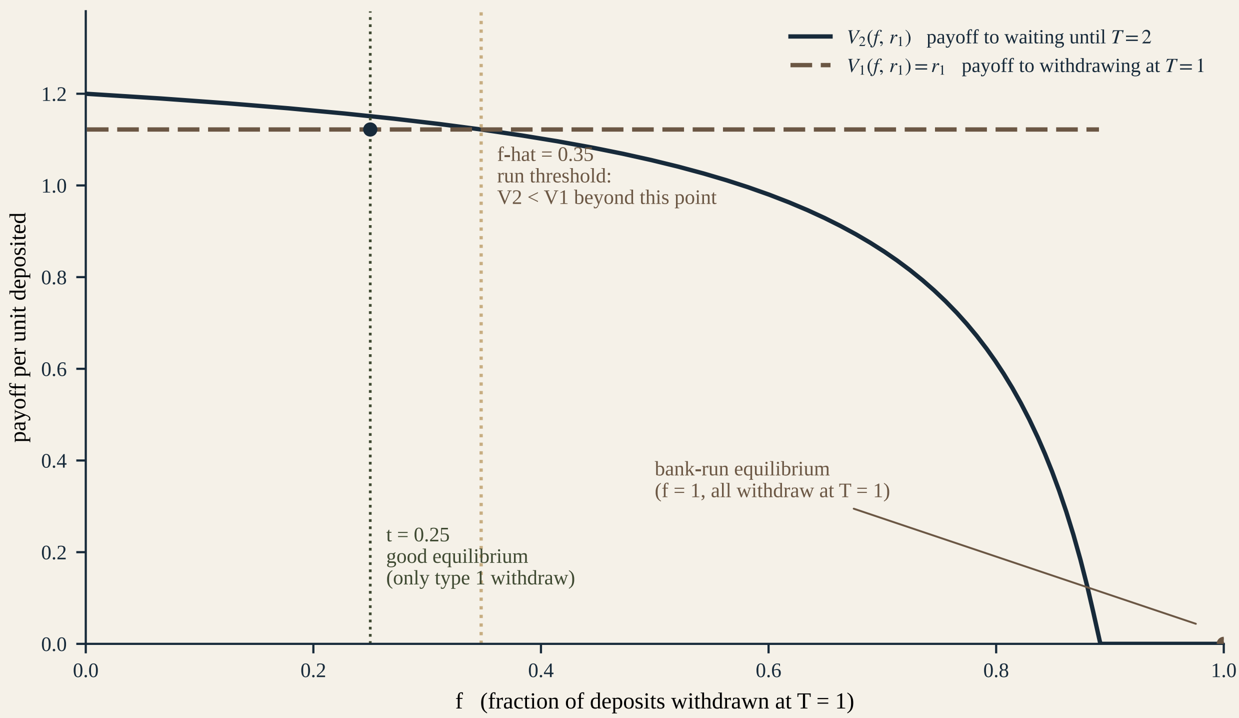

Figure 2. Withdrawal payoffs $V_1(f, r_1)$ and $V_2(f, r_1)$ as functions of the aggregate withdrawal fraction f

Source: reviewer's calculation from equations (2)–(3), $r_1 = c_1^* \approx 1.122$, R = 1.20. Not contained in the original paper. The horizontal dashed line at r_1 is the constant early-withdrawal payoff up to the capacity limit $f = 1/r_1$; the declining curve is the late-withdrawal payoff, which crosses below r_1 at $\bar{f} \approx 0.35$ in this calibration — beyond that point, waiting is dominated and withdrawing immediately is a best response for every agent, which is the mechanical origin of the run equilibrium at f = 1. The relationship plotted is exact (it follows directly from the definitions of V_1 and V_2), not estimated, so there are no residuals, outliers, or nonlinearities beyond the single kink at $f = 1/r_1$ built into V_1 by construction.

Two features of Figure 2 are worth isolating analytically. First, the crossing point f is strictly greater than t in this calibration (0.35 versus 0.25), which is what makes $f = t$ (the good equilibrium) a locally stable-looking outcome even though it coexists with the run: a small, idiosyncratic deviation by a handful of type 2 agents does not by itself trigger a cascade, because V_2 still exceeds V_1 in the neighborhood of $f = t$. Second, and more importantly, nothing in the model determines whether the economy coordinates on $f = t$ or $f = 1$; both are equally valid Nash equilibria of the identical game, and the figure cannot by construction show which one obtains, because that is exactly the equilibrium-selection question the paper leaves open (see §7 and §8).

6. Robustness and Additional Tests

CONCERN: THE DEMAND-DEPOSIT CONTRACT MIGHT BE A FRAGILE ARTIFACT OF A PARTICULAR FUNCTIONAL FORM

Test — the paper allows V_1 and V_2 to be *any* functions of queue position and aggregate withdrawals, not only the specific linear form in (2)–(3), when proving Proposition 1. **Result —** the impossibility result holds for the entire class of sequential-service contracts, not merely the simple demand-deposit contract. **Does it resolve the concern?** Yes, on its own terms: this is a genuine generality check, and it is one of the paper's more careful pieces of proof technique, elevating Proposition 1 from a statement about one contract to a statement about an entire mechanism class.

CONCERN: SUSPENSION OF CONVERTIBILITY MIGHT JUST BE A RELABELED VERSION OF THE RUN EQUILIBRIUM IT IS MEANT TO PREVENT

Test — the authors verify that under the suspension contract, waiting strictly dominates withdrawing at $T = 1$ for every $f_j \leq f$, regardless of what an agent believes about others' actions. **Result —** this is a dominant-strategy argument, not merely a best-response-given-correct-beliefs argument, which is the strongest equilibrium concept used anywhere in the paper. **Does it resolve the concern?** Yes, conditional on t being deterministic; the authors are explicit that this conditionality is exactly what Proposition 1 then removes.

CONCERN: IS THE DEPOSIT-INSURANCE RESULT REALLY ABOUT INSURANCE, OR IS IT EQUIVALENT TO A LENDER-OF-LAST-RESORT DISCOUNT WINDOW?

Test — the conclusion argues verbally that a discount window purchasing bank assets at $T = 1$ with money-creation tax revenue, calibrated identically to the optimal deposit-insurance tax, would have the same effect under the riskless technology assumed throughout the paper. **Result —** asserted equivalence under riskless technology; explicitly *not* re-derived formally, and the authors state that the equivalence breaks down once the technology is risky, because a discount window that is always expected to bail out illiquid banks creates an incentive to take on interest-rate or credit risk ex ante. **Does it resolve the concern?** No — this is the one place in the paper where a robustness claim is made and then immediately qualified away without a model to back it up. CRITIQUE Treating this as a settled robustness check would overstate what the paper shows; it is better read as a research agenda item the authors correctly flag but do not pursue.

CONCERN: DOES THE MECHANISM GENERALIZE BEYOND BANKS NARROWLY DEFINED?

Test — the conclusion discusses, verbally, a non-bank firm funded with short-term debt facing the same rollover-refusal coordination problem, and notes that bankruptcy protection functions analogously to suspension of convertibility. **Result —** a plausible analogy, supported by the empirical observation (also asserted, not modeled) that banks hold a large share of corporate short-term debt and that commercial-paper issuers commonly maintain backup bank lines of credit. **Does it resolve the concern?** Partially, as a plausibility argument; it correctly anticipates a research direction (rollover risk in corporate finance, later formalized by, among others, He and Xiong 2012 on dynamic debt runs) but is not itself a formal robustness test within this paper.

7. Critical Evaluation

7.1 What the Paper Actually Establishes

> **ESTABLISHED**
>
> A demand-deposit-style contract subject to sequential service can implement full-information-optimal risk sharing as one equilibrium when the fraction of early consumers is deterministic (rigorously proved). The identical contract admits a strictly Pareto-worse, self-fulfilling run equilibrium for any early-withdrawal value above the technology's liquidation value (rigorously proved). No sequential-service-constrained contract of any functional form can achieve full efficiency once aggregate withdrawal demand is stochastic (Proposition 1, rigorously proved by contradiction).

> **SUPPORTED BUT CONDITIONAL**
>
> Suspension of convertibility eliminates the run equilibrium and restores the optimum as a dominant-strategy equilibrium — conditional on the bank knowing the true withdrawal demand t in advance and being able to commit credibly to the threshold $\hat{f}$. Government deposit insurance achieves the unconstrained optimum uniquely (Proposition 2) — conditional on the government's taxing power being costless, unconstrained, and capable of conditioning on the full realized withdrawal path, an assumption the authors themselves call potentially too strong.

> **SUGGESTIVE**
>
> The claimed equivalence between deposit insurance and a discount-window lender of last resort (asserted only under riskless technology, not modeled). The extension of the illiquid-asset/liquid-liability logic to non-bank firms financed with short-term debt (discussed only informally in the conclusion).

> **NOT ESTABLISHED**
> Why agents in fact coordinate on the run equilibrium rather than the good one — the model is silent on the probability or trigger of a run and treats the selection between multiple equilibria as exogenous ("sunspots," in the authors' own word, citing Azariadis 1980 and Cass and Shell 1983). Any bound on the welfare cost of a run relative to alternative institutional arrangements. Behavior under a risky (rather than riskless) technology, which the authors explicitly flag as unmodeled. Any account of a banking system with more than one bank, interbank linkages, or contagion across institutions — the single representative bank stands in for the entire intermediation sector throughout.

7.2 Strengths

Recognized by the authors: the model gives banks an explicit economic function — transforming a partially illiquid technology into liquid claims under private information — that Arrow-Debreu markets cannot replicate; it explains, rather than merely assumes, why deposit insurance can be welfare-improving beyond correctly pricing an exogenous default probability, in contrast to Merton (1977, 1978) and Kareken and Wallace (1978).

Identified independently by this review: the sharp separation between what a privately feasible, sequentially-serving contract can achieve and what an entity with ex post, economy-wide conditioning power can achieve is a genuine mechanism-design insight that reaches well beyond banking — it is a clean illustration of how a seemingly small institutional detail (the order in which claims are honored) can be the entire source of an equilibrium multiplicity. The model is also strikingly parsimonious: three dates, one good, one technology, and no aggregate uncertainty are sufficient to generate the whole multiplicity result, which is a demonstration of theoretical economy that later, richer extensions (global games, multi-bank contagion) have had to work considerably harder to match in tractability. Finally, treating runs as a Nash equilibrium object, rather than as an exogenously specified failure probability, set the entire subsequent research agenda in banking theory, including refinements the authors themselves could not have anticipated (unique-equilibrium selection via global games, discussed in §9).

7.3 Weaknesses

THE RUN TRIGGER IS EXOGENOUS AND UNMODELED

Problem — the model proves a bad equilibrium exists but offers no theory of what causes agents to coordinate on it rather than the good one; the authors themselves attribute coordination to "almost anything," including sunspots. **Why it matters** — without a theory of the trigger, the model has no predictive content about when runs occur, only that they can occur; this materially limits its usefulness for calibrating the probability of a crisis or for designing policy aimed at reducing that probability rather than only its consequences. **Potential bias** — the indeterminacy makes the model compatible, after the fact, with almost any observed pattern of runs, which weakens its falsifiability as a positive theory even as its normative logic (suspension, insurance) remains sound. **How it could be tested** — the global-games refinement (Morris and Shin 1998; formalized for this exact model by Goldstein and Pauzner 2005) perturbs the common-knowledge assumption with a small amount of private, correlated noise about fundamentals, which is enough to select a unique equilibrium and to generate a run probability that is a genuine, testable function of the model's parameters.

THE TECHNOLOGY IS RISKLESS

Problem — R is a known constant; the model has no state in which the bank's assets are genuinely impaired. **Why it matters —** the paper explicitly positions itself against Fisher's (1911) insolvency-based view of runs, yet by construction cannot represent that view even as a special case, so it cannot speak to the empirically important question of how much of historical run activity was panic-driven versus fundamentals-driven — a distinction the later literature (Gorton 1988; Chari and Jagannathan 1988) treats as first-order. **Potential bias —** using the model uncritically for policy calibration risks attributing to "pure panic" costs that, in a given historical episode, were partly or largely fundamentals-driven. **How it could be tested —** introduce asset-side risk explicitly (Allen and Gale 1998) and check whether the coordination-failure channel and the fundamentals channel can be separately identified from observable withdrawal and asset-quality data.

DEPOSIT INSURANCE IS OPTIMAL ONLY UNDER COSTLESS, UNCONSTRAINED TAXATION

Problem — Proposition 2's tax scheme requires the government to levy a state-contingent, ex post tax on withdrawn wealth with no distortion and no limit. **Why it matters —** real deposit insurance is financed by flat, ex ante premia, not state-contingent ex post levies, so the specific financing instrument that makes Proposition 2 work is not the instrument regulators actually use; the authors concede the result "may be too strong." **Potential bias —** citing Proposition 2 as a general efficiency case for public deposit insurance, without this caveat, overstates what has actually been shown. **How it could be tested —** replace the ex post lump-sum tax with a realistic ex ante premium schedule, or add collection costs and distortions, and check whether uniqueness and full efficiency survive; this is essentially the research program the deposit-insurance-pricing literature pursued in the following decade.

A SINGLE BANK STANDS FOR THE ENTIRE BANKING SYSTEM

Problem — there is no interbank market, no heterogeneity across banks, and no channel for a run on one institution to affect beliefs about another. **Why it matters —** contagion across nominally unrelated institutions is a first-order feature of every major historical banking crisis the paper itself invokes (the 1930s panics, and by extension the 2008 and 2023 episodes it could not have anticipated). **Potential bias —** policy conclusions drawn from a single-bank model may understate the case for system-wide interventions (stress tests, resolution regimes) relative to single-institution ones (deposit insurance, suspension). **How it could be tested —** multi-bank extensions with correlated signals about individual bank quality, of the kind developed later in the contagion literature.

MORAL HAZARD ON THE ASSET SIDE IS RAISED BUT NOT MODELED

Problem — the conclusion notes that if bank managers can choose portfolio risk unobserved, deposit insurance could distort that choice, but this channel is never formalized. **Why it matters —** this is exactly the risk-shifting mechanism that became central to the U.S. savings-and-loan crisis unfolding contemporaneously with the paper's publication, which the introduction itself references; the paper therefore recommends an instrument (deposit insurance) whose principal real-world cost it explicitly identifies but does not size. **How it could be tested —** combine the Diamond-Dybvig run mechanism with an endogenous-portfolio-choice model in the spirit of Kareken and Wallace (1978).

> **SINGLE MOST IMPORTANT WEAKNESS**
>
> Of these, the unmodeled equilibrium-selection trigger is the most consequential, because it is not a peripheral extension but a gap at the center of the paper's own explanatory claim. A theory that shows a bad equilibrium *can* exist, without saying anything about when or why an economy lands on it, cannot by itself distinguish a world in which runs are a live, first-order risk from one in which they are a theoretical curiosity that never actually binds — and this is precisely the gap that thirty subsequent years of research on equilibrium selection in the model (Postlewaite and Vives 1987; Chari and Jagannathan 1988; Peck and Shell 2003; Goldstein and Pauzner 2005; Rochet and Vives 2004) exists to close.

8. Contradictions and Edge Cases

8.1 When Does the Liquidity-Insurance Result Reverse Sign?

The paper assumes relative risk aversion strictly greater than one everywhere as a sufficient condition for $c_1^* > 1$ and $c_2^* < R$. Using the same CRRA calibration as Table 1 ($R = 1.20$, $\rho = 0.90$, $t = 0.25$), the reviewer solved the model at successively lower values of γ to test how tight this sufficient condition actually is.

Table 2. Sign reversal of the liquidity-insurance result as relative risk aversion falls

γ (relative risk aversion)	c_1*	c_2*	Sign matches paper's prediction?
6.00	1.132	1.147	Yes (c_1*>1, c_2*<R)
3.00	1.122	1.151	Yes
1.00	1.081	1.168	Yes
0.50	1.022	1.191	Yes
0.4221 (crossing point)	1.000	1.200	Exactly autarky — no insurance value
0.20	0.856	1.258	**No** — sign reversed

Source: reviewer's calculation, solving (1b)–(1c) for CRRA utility at each γ, holding R, ρ, t fixed at the Table 1 calibration.

EXTENSION Two findings follow from this exercise, and neither is stated in the paper. First, the paper's sufficient condition ($\gamma > 1$) is not tight: in this calibration the qualitative result survives down to $\gamma \approx 0.42$, comfortably below the stated threshold, which means the assumption as written is conservative rather than necessary. Second, and more important economically, below that crossing point the sign genuinely reverses: the "optimal" contract would pay early consumers *less* than the liquidation value of their own claim and late consumers *more* than the technology's full return, which is not a liquidity-insurance contract in any meaningful sense — no rational depositor would accept $r_1 < 1$ in exchange for keeping money in a demand deposit, since holding cash (or the equivalent of the liquidation value) directly dominates it. This edge case shows that the entire liquidity-transformation rationale for banking in this model depends on agents being sufficiently risk averse, in a specific and falsifiable quantitative sense the paper does not itself quantify.

8.2 Extreme Values of the Withdrawal Fraction

As $t \to 0$, almost no one is an early consumer, so the aggregate insurance benefit (measured in total welfare) shrinks even though the per-person benefit to the rare type-1 agent remains large; as $t \to 1$, almost everyone is an early consumer, and $c_1^* \to 1$ because there is too little late-consumer wealth left to redistribute — the insurance motive collapses at both extremes for the same underlying reason, that gains from pooling require a genuine mix of types in the realized population, not merely idiosyncratic risk ex ante.

8.3 Could the Same Observed Behavior Arise from a Different Mechanism?

Mass withdrawal from a bank is observationally similar whether it is a Diamond-Dybvig coordination failure or an information-based run in which depositors rationally infer, from a public signal, that the bank's assets are genuinely impaired (Jacklin and Bhattacharya 1988; Chari and Jagannathan 1988). The two have opposite policy implications: deposit insurance eliminates the former (Proposition 2) but does nothing to fix — and may even worsen, via moral hazard — the latter, since an insolvent bank does not become solvent by insuring its depositors. This paper, taken alone, cannot distinguish the two from withdrawal data; doing so requires either the additional structure Jacklin and Bhattacharya introduce or the fundamentals-based equilibrium selection of Peck and Shell (2003) and Goldstein and Pauzner (2005).

8.4 The Sequential-Service Constraint Under Instant Digital Withdrawal

The entire logic of suspension of convertibility, and the entire distinction between an individual's queue position and the aggregate withdrawal rate, presumes a meaningful processing lag: the bank observes withdrawals arriving one at a time and can, in principle, act on that information before everyone has been served. Modern retail and wholesale banking, in which withdrawal and transfer instructions are executed electronically and essentially simultaneously across tens of thousands of depositors, compresses this lag toward zero. EXTENSION This is not a modification of the model's parameters but a modification of its core institutional assumption, and it cuts in a direction the model itself predicts should matter: if the time available to observe f and impose suspension before the queue empties shrinks toward zero, the suspension-based defense analyzed in §3.7 loses the window in which it operates, even though the underlying multiple-equilibrium structure is unchanged. The March 2023 run on Silicon Valley Bank, in which a very large share of a concentrated, largely uninsured deposit base was withdrawn or requested for withdrawal within a single business day via electronic transfer, is consistent with exactly this reading: not a new economic mechanism relative to Diamond and Dybvig's model, but the same mechanism operating at a queue-processing speed the 1983 institutional environment did not contemplate. This edge case is developed further as a research opportunity in §11.

9. Further Literature

PRECEDED IT

Bryant (1980) — A Model of Reserves, Bank Runs, and Deposit Insurance. Shared idea: reserves, withdrawal risk, and deposit insurance in a single reduced-form banking model. Relationship: an acknowledged direct predecessor, cited in the paper's introduction; Diamond and Dybvig supply the fully specified, incentive-compatible mechanism-design derivation that Bryant's more reduced-form treatment lacks. Why it matters: establishes that the core intuition was independently emerging in the literature circa 1980, and that this paper's distinct contribution is formalization and generality, not the base intuition itself.

ESTABLISHED THE THEORETICAL FOUNDATION

Azariadis (1980) and **Cass and Shell (1983) —** self-fulfilling prophecies and "sunspot" equilibria. Shared idea: extrinsic, non-fundamental variables can determine which of multiple rational-expectations equilibria obtains. Relationship: Diamond and Dybvig explicitly invoke this apparatus to explain what selects the run equilibrium, without further elaborating it themselves. Why it matters: this is the theoretical vocabulary the paper borrows to avoid having to explain equilibrium selection on its own terms — and is exactly the gap later closed by global-games refinements.

DEVELOPED SIMILAR IDEAS / COMPETING ACCOUNTS OF RUNS

Fisher (1911) — runs as a rational response to a decline in asset value below fixed nominal liabilities. Relationship: the paper explicitly positions its coordination-failure account against this insolvency-based view. Why it matters: the two views are not mutually exclusive empirically, and distinguishing them (Gorton 1988; Chari and Jagannathan 1988) became a central empirical research question the theory alone cannot settle.

Kareken and Wallace (1978) and **Merton (1977, 1978) —** deposit insurance pricing under an exogenous failure probability. Relationship: predecessor literature the paper explicitly contrasts itself with, on the grounds that pricing insurance against an assumed failure probability is a different question from explaining why insurance is welfare-improving. Why it matters: clarifies exactly what is new in Diamond-Dybvig's treatment of insurance — the welfare rationale, not the pricing mechanics.

CONTRADICTED OR QUALIFIED ITS FINDINGS

Wallace (1988) — "Another Attempt to Explain an Illiquid Banking System: The Diamond and Dybvig Model with Sequential Service Taken Seriously," Federal Reserve Bank of Minneapolis Quarterly Review. Shared idea: takes the sequential-service constraint even more literally than the original paper and shows that doing so changes some of the model's implications. Relationship: an internal critique from within the same modeling tradition. Why it matters: shows the results are more sensitive to the precise formalization of sequential service than the original exposition suggests.

Jacklin (1987) — Demand Deposits, Trading Restrictions, and Risk Sharing. Shared idea: asks whether the same risk-sharing allocation can be achieved through tradeable equity claims rather than demand deposits. Relationship: shows that under some conditions an equity-like claim can achieve the same risk sharing without the run vulnerability, which qualifies the paper's implicit suggestion that demand deposits are the only, or the best, instrument for this purpose. Why it matters: reframes the run vulnerability as a property of the specific contract chosen, not an unavoidable consequence of providing liquidity insurance at all.

Jacklin and Bhattacharya (1988) — Distinguishing Panics and Information-Based Bank Runs. Shared idea: extends the framework to allow a public signal about asset quality. Relationship: shows the coordination-failure and insolvency-based accounts of runs (Diamond-Dybvig versus Fisher) can be nested in one model and empirically distinguished. Why it matters: directly operationalizes the distinction raised in §8.3 above.

IMPROVED ITS IDENTIFICATION OF EQUILIBRIUM SELECTION

Postlewaite and Vives (1987) — Bank Runs as an Equilibrium Phenomenon. Shared idea: endogenizes the run probability from fundamentals rather than leaving it as an unmodeled sunspot. **Chari and Jagannathan (1988) —** Banking Panics, Information, and Rational Expectations Equilibrium. Shared idea: depositors use the size of withdrawal queues as a noisy signal of bank solvency, generating information-based runs. **Peck and Shell (2003) —** Equilibrium Bank Runs. Shared idea: runs can be driven by fundamentals within a mechanism-design-consistent version of the original model, without appeal to sunspots. **Goldstein and Pauzner (2005) —** Demand-Deposit Contracts and the Probability of Bank Runs. Shared idea: applies global-games techniques to the Diamond-Dybvig contract, yielding a unique equilibrium and a run probability that is an explicit, comparative-statics function of the contract's parameters. **Rochet and Vives (2004) —** Coordination Failures and the Lender of Last Resort. Shared idea: extends the global-games logic to lender-of-last-resort policy design. Why these matter collectively: this cluster of papers is the direct answer to the single most important weakness identified in §7.3 — the absence of an equilibrium-selection theory in the original model.

EXTENDED IT STRUCTURALLY

Green and Lin (2000) — "Diamond and Dybvig's Classic Theory of Financial Intermediation: What's Missing?" Federal Reserve Bank of Minneapolis Quarterly Review. Shared idea: shows that with a finite number of agents and full information about the realized sequence of types, the run equilibrium can sometimes be eliminated by a more careful mechanism, without appeal to government intervention. Relationship: a technical refinement of exactly how much of the multiplicity result survives under alternative informational assumptions. **Allen and Gale (1998) —** Optimal Financial Crises. Shared idea: reintroduces aggregate (not just idiosyncratic) risk into the framework, showing that some crises can be constrained-efficient responses to aggregate shocks rather than pure coordination failures. **Diamond and Rajan (2001) —** Liquidity Risk, Liquidity

Creation, and Financial Fragility: A Theory of Banking. Shared idea: connects the liquidity-transformation role of banks to their lending/monitoring role, unifying this paper's account with Diamond's (1984) delegated-monitoring theory.

APPLIED IT EMPIRICALLY OR TO NEW SETTINGS

Gorton (1988) — Banking Panics and Business Cycles. Empirically tests whether pre-1914 U.S. banking panics were better predicted by macroeconomic fundamentals (consistent with an information-based or insolvency view) or by panic indicators independent of fundamentals, using U.S. National Banking Era data — the empirical companion the original theory paper does not itself provide. **Calomiris and Kahn (1991) —** The Role of Demandable Debt in Structuring Optimal Banking Arrangements. Applies a disciplining, rather than purely fragile, interpretation of demand deposits, in which the threat of withdrawal disciplines bank management — an alternative rationale for the same contractual form. **Gorton and Metrick (2012) —** Securitized Banking and the Run on Repo. Applies the run logic to short-term wholesale funding (repo) in the 2007–2008 crisis, well outside the retail-deposit setting of the original paper. **Jiang, Matvos, Piskorski, and Seru (2023) —** work on U.S. bank fragility and uninsured deposits during the March 2023 regional-bank turmoil, applying the uninsured-deposit-run logic of this paper directly to the Silicon Valley Bank episode discussed in §8.4.

THE AUTHORS' OWN FOLLOW-UP

Diamond and Dybvig (1986) — Banking Theory, Deposit Insurance, and Bank Regulation, Journal of Business. A policy-oriented restatement of the 1983 model's implications for bank regulation, written by the same authors for a practitioner-facing audience.

10. Research Gaps

10.1 Paper-Level Gaps

Gap — no theory of equilibrium selection or run probability. **Why it matters —** without it, the model cannot generate testable comparative statics on when runs occur, only that they can. **Existing limitation —** the paper attributes selection to unmodeled sunspots. **Possible way forward —** global-games refinement (already substantially pursued by Goldstein and Pauzner 2005; still open for extension to richer contract spaces).

Gap — riskless technology precludes any role for genuine insolvency. **Why it matters —** the empirically important panic-versus-fundamentals distinction cannot be addressed within the model as written. **Existing limitation —** R is a known constant throughout. **Possible way forward —** Allen and Gale's (1998) aggregate-risk extension, and further work nesting Fisher's (1911) insolvency channel and the coordination-failure channel in one estimable framework.

Gap — Proposition 2's financing instrument (costless, unconstrained, ex post lump-sum taxation) does not match real deposit-insurance financing (ex ante flat premia). **Why it matters —** the efficiency case for public deposit insurance, as actually implemented, has not been shown by this paper on its own terms. **Existing limitation —** the authors concede the assumption may be too strong. **Possible way forward —** re-derive Proposition 2 under realistic, distortionary, ex ante financing and characterize how much of the efficiency gain survives.

10.2 Literature-Level Gaps

Forty years on, the literature has still not fully settled the empirical division of historical and modern banking crises into panic-driven and fundamentals-driven components; Gorton (1988) and its successors offer partial, era-specific answers, but a general, out-of-sample-validated method for making this classification in real time (as opposed to after the fact) remains an open problem with direct relevance to supervisory policy. Similarly unresolved is how the classic sequential-service/suspension logic should be re-derived for an environment of instantaneous electronic withdrawal, where the processing-time assumption that both the good-equilibrium result and the suspension defense implicitly rely on no longer holds in anything like its 1983 form.

10.3 Methodological Gaps

Better **data** would mean high-frequency, depositor-level withdrawal data around actual run episodes (increasingly available for 2023-era events given electronic transfer records) rather than the aggregate, low-frequency data used in most of the historical empirical literature. Better **identification** would mean natural experiments that shift depositors' beliefs about aggregate withdrawal behavior without simultaneously shifting beliefs about fundamentals, which is difficult because public signals about a bank typically move both simultaneously. Better **models** would mean tractable multi-bank extensions with correlated, privately observed signals about individual bank quality, capable of speaking to contagion, which the original single-bank framework structurally cannot address.

11. Research Opportunities

Idea 1 — Withdrawal velocity and the collapse of the sequential-service defense in digital banking

RESEARCH QUESTION: Has the diffusion of instant electronic withdrawal and transfer compressed the time window over which suspension-of-convertibility-style defenses can operate, and does this measurably raise realized run severity conditional on a run beginning?

MOTIVATION: §8.4 above — the entire suspension mechanism in §3.7 presumes a processing lag that modern mobile and wire transfer largely eliminates.

HYPOTHESIS: Banks with a higher share of depositors using mobile/API-based withdrawal channels exhibit faster peak-to-trough deposit outflow during a run episode, controlling for deposit insurance coverage and bank size.

ECONOMIC MECHANISM: Direct test of the sequential-service constraint's empirical bite, per §3.5–3.7.

LITERATURE CONNECTION: Diamond and Dybvig (1983); Jiang, Matvos, Piskorski, and Seru (2023).

DATA: FDIC call reports; high-frequency deposit-outflow data disclosed in post-mortem regulatory reports (e.g., the Federal Reserve's and FDIC's own reviews of the March 2023 failures); comparison sample of pre-2010 run episodes without mobile banking.

VARIABLES: hourly/daily deposit outflow rate; share of digitally-active accounts; uninsured deposit share; bank size and concentration.

METHOD: event-study comparison of outflow velocity across the pre- and post-mobile-banking eras, with bank and time fixed effects.

IDENTIFICATION STRATEGY: staggered adoption of mobile/API withdrawal channels across banks as a source of variation in withdrawal technology, holding deposit-insurance regime fixed.

EXPECTED CONTRIBUTION: the first direct empirical test of whether the sequential-service constraint's bite has weakened over time, connecting a 1983 theoretical assumption to a 2020s institutional fact.

MAIN RISK: confounding between digital-banking adoption and the type of bank (e.g., technology-sector-concentrated deposit bases) that is independently more run-prone for reasons unrelated to withdrawal speed.

Idea 2 — Uninsured deposit concentration and run probability: a cross-sectional test of the model's central comparative static

RESEARCH QUESTION: Does the cross-sectional share of uninsured deposits at U.S. banks predict run incidence and severity during periods of monetary tightening, as Proposition 2 implies it should (insured deposits should not run; uninsured deposits remain exposed to the coordination failure of §3.6)?

MOTIVATION: a direct empirical test of the paper's core policy claim — that deposit insurance, not merely bank quality, is what removes run vulnerability.

HYPOTHESIS: conditional on measured asset-side fundamentals, banks with a higher uninsured deposit share experience larger deposit outflows following a common shock to depositor confidence.

ECONOMIC MECHANISM: §3.9 — insured deposits are effectively guaranteed the model's $\tilde{V}_1(f) \leq \tilde{V}_2(f)$ property; uninsured deposits are not.

LITERATURE CONNECTION: Diamond and Dybvig (1983) Proposition 2; Jiang, Matvos, Piskorski, and Seru (2023).

DATA: FDIC quarterly call reports (uninsured deposit share is a reported field), 2018–2024 panel, spanning the 2023 tightening episode.

VARIABLES: uninsured deposit share; deposit growth rate; unrealized securities losses (mark-to-market fundamentals control); bank size.

METHOD: panel regression with bank and quarter fixed effects; heterogeneity analysis by fundamentals quartile to separate the pure-insurance channel from a fundamentals channel.

IDENTIFICATION STRATEGY: difference-in-differences around the March 2023 shock, comparing high- versus low-uninsured-share banks with similar measured fundamentals.

EXPECTED CONTRIBUTION: a direct, modern-data test of the paper's central normative claim, forty years after publication.

MAIN RISK: uninsured deposit share is plausibly correlated with unobserved bank business-model risk (e.g., concentrated depositor base), which would bias the estimated insurance effect.

Idea 3 — A global-games test of run probability using the 2023 U.S. regional-bank episode

RESEARCH QUESTION: Does the Goldstein-Pauzner (2005) global-games extension of this model generate run-probability predictions that fit the cross-section of which regional banks did and did not experience a run in March 2023?

MOTIVATION: tests whether the equilibrium-selection refinement that answers this paper's central open question (§7.3, §9) has out-of-sample empirical content.

HYPOTHESIS: banks whose estimated fundamentals placed them inside the global-game "region of doubt" (near the boundary where solvency is genuinely uncertain to depositors) were disproportionately likely to experience a run, relative to banks clearly inside or outside that region.

ECONOMIC MECHANISM: global games replace the sunspot-selection gap in §7.3 with a noise-driven unique equilibrium; the model generates an explicit run-probability function of fundamentals and noise dispersion.

LITERATURE CONNECTION: Goldstein and Pauzner (2005); Morris and Shin (1998); Diamond and Dybvig (1983).

DATA: bank-level mark-to-market losses on held-to-maturity securities portfolios (as estimated by Jiang, Matvos, Piskorski, and Seru 2023); deposit outflow data; equity market signals as a proxy for the private noisy signal in the global-games structure.

VARIABLES: estimated mark-to-market capital shortfall; deposit insurance coverage ratio; realized run indicator.

METHOD: structural estimation of the global-games run-probability function, followed by an out-of-sample classification exercise against realized 2023 outcomes.

IDENTIFICATION STRATEGY: cross-sectional variation in estimated fundamentals shortfall across banks facing the same aggregate interest-rate shock.

EXPECTED CONTRIBUTION: the first structural, out-of-sample test of the global-games refinement against a real, well-documented run episode.

MAIN RISK: the global-games model is highly stylized and its mapping from observable fundamentals to the theoretical "region of doubt" requires strong auxiliary assumptions that are themselves untested.

Idea 4 — Money market fund runs as a Diamond-Dybvig setting without deposit insurance

RESEARCH QUESTION: Do prime money market fund runs (2008 Reserve Primary Fund; 2020 COVID-period prime fund outflows) fit the Diamond-Dybvig equilibrium structure, and does the absence of Proposition-2-style insurance explain their persistence relative to insured bank deposits?

MOTIVATION: money market funds perform near-identical liquidity transformation to banks but historically lacked an equivalent backstop, offering a natural test of whether the insurance channel, specifically, is what stabilizes insured deposits.

HYPOTHESIS: fund-level outflow severity during 2008 and 2020 is decreasing in the fund's effective access to a backstop (Federal Reserve facilities, sponsor support) and increasing in redemption-queue sensitivity analogous to the model's sequential-service constraint.

ECONOMIC MECHANISM: §3.6–3.9, applied to a non-bank liquidity-transformation vehicle.

LITERATURE CONNECTION: Diamond and Dybvig (1983); Gorton and Metrick (2012) on repo runs as a structurally similar case.

DATA: SEC Form N-MFP filings (daily/weekly fund-level flow and portfolio data), 2008 and 2020 episodes.

VARIABLES: daily net redemptions; portfolio liquidity composition; sponsor-support indicator; access to Fed facilities.

METHOD: panel event study around each crisis episode with fund fixed effects.

IDENTIFICATION STRATEGY: cross-sectional variation in backstop access among funds facing the same aggregate liquidity shock.

EXPECTED CONTRIBUTION: extends the paper's insurance-versus-no-insurance comparative static to a non-bank setting with high-quality regulatory data.

MAIN RISK: money market funds differ from the model's bank in ways beyond insurance (e.g., mark-to-market NAV rules, gates and fees introduced after 2016), which could confound the comparison.

Idea 5 — Stablecoin redemption dynamics as a new asset-class test of the model

RESEARCH QUESTION: Do stablecoin de-pegging and redemption episodes (Terra/UST 2022; USDC's brief de-peg following Silicon Valley Bank's failure in March 2023) display the same multiple-equilibrium structure as the demand-deposit contract analyzed in §3.6, and does on-chain settlement (near-instant, transparent, but still sequential at the smart-contract level) alter the suspension-of-convertibility logic of §3.7?

MOTIVATION: stablecoins are a genuinely new instance of illiquid-asset-backed, on-demand-redeemable liabilities, and on-chain data offers a transparency the original model's bank setting never had.

HYPOTHESIS: redemption velocity and the presence or absence of a credible backstop (reserve transparency, issuer support) predict whether a de-pegging episode resolves toward the "good" equilibrium (peg restored) or the "run" equilibrium (continued redemption at a discount).

ECONOMIC MECHANISM: §3.6, reinterpreted with the smart-contract redemption queue as the sequential-service constraint and on-chain reserve attestations as a (partial) substitute for deposit insurance.

LITERATURE CONNECTION: Diamond and Dybvig (1983); the algorithmic-stablecoin literature emerging after the Terra/UST collapse.

DATA: on-chain transaction and redemption data (publicly observable by construction for most major stablecoins), exchange-level price deviation from peg.

VARIABLES: redemption volume and velocity; reserve composition and transparency; peg deviation.

METHOD: event studies around de-pegging episodes; cross-sectional comparison of stablecoins with different reserve-backing and transparency regimes.

IDENTIFICATION STRATEGY: cross-sectional variation in reserve transparency and backstop credibility across stablecoins facing correlated market stress (e.g., the March 2023 banking-sector shock, which affected multiple stablecoin issuers' banking relationships simultaneously).

EXPECTED CONTRIBUTION: the first application of the Diamond-Dybvig apparatus, with its original policy conclusion (insurance/backstop credibility restores uniqueness), to a genuinely new, transparent asset class.

MAIN RISK: on-chain data availability is uneven across issuers, and reserve composition disclosures are self-reported and not independently audited at the same standard as bank call reports, which is itself a data-quality note that should be stated explicitly rather than estimated around.

Ranking

Ranked on novelty, feasibility, data availability, methodological quality, economic importance, and potential contribution, the strongest three ideas are, in order:

1. Idea 2 (uninsured deposit concentration and run probability). Highest feasibility and data quality — FDIC call reports are a clean, quarterly, mandatorily reported panel with the exact variable (uninsured deposit share) the theory identifies as the key comparative static, and the March 2023 episode provides a sharply dated natural experiment. It is the most direct test of the paper's own central claim.

2. Idea 1 (withdrawal velocity and digital banking). High novelty and direct connection to a genuine gap in the original model's institutional assumptions (§8.4); feasibility is somewhat lower because granular, hourly deposit-outflow data is not routinely disclosed and depends on regulatory post-mortem reports being sufficiently detailed.

3. Idea 3 (global-games test using 2023 data). Highest methodological ambition and the most direct engagement with the single most important weakness identified in §7.3, but feasibility is constrained by the auxiliary structural assumptions the global-games mapping requires, which are themselves difficult to validate independently.

Ideas 4 and 5 are promising but rank lower here on data quality grounds: money market fund flow data (Idea 4) is good but the setting differs from the model along several dimensions simultaneously, complicating clean identification; stablecoin data (Idea 5) is novel and topical but self-reported reserve disclosures currently fall short of the audit standard available for regulated banks, a limitation that should be stated as such rather than assumed away.

12. Learning Framework

Prerequisite Concepts

Expected-utility theory under uncertainty; Nash equilibrium and the stronger concept of a dominant-strategy equilibrium; constant relative risk aversion and the economic meaning of the risk-aversion coefficient; Arrow-Debreu contingent-claims markets and why they fail under private, non-verifiable information; incentive compatibility and self-selection constraints; the basic economics of maturity and liquidity transformation.

Core Concepts Learned

That banks can be derived, rather than assumed, as a response to a specific informational friction (privately observed liquidity type); that the same contract that performs a genuine risk-sharing service can simultaneously admit a self-fulfilling, welfare-reducing equilibrium, and that these two properties are linked rather than separable; that the sequential-service constraint is the single structural feature responsible for that link; and that a government's ability to condition transfers on ex post, economy-wide information — not merely its financial capacity — is what can, in principle, break the link.

Important Equations

The first-order condition for optimal risk sharing, $u'(c_1^*) = \rho R \cdot u'(c_2^*)$, worth remembering as a general template for any intertemporal consumption-smoothing problem; the sequential-service payoff functions V_1 and V_2 in equations (2)–(3), worth remembering as the canonical way to formalize a first-come-first-served constraint inside a Nash-equilibrium framework; and the resource constraint $t \cdot c_1^* + (1-t) \cdot c_2^*/R = 1$, worth remembering as the general form linking a liquidation-cost technology to a planner's feasible allocation set.

Methods Learned

Proof-by-contradiction impossibility results in mechanism design (the technique underlying Proposition 1); the use of dominant-strategy equilibrium, rather than merely Nash equilibrium, as a stability concept when credibility of commitment is itself in question; the general technique of asking what an institution structurally can and cannot condition its payments on (queue position versus realized aggregate outcome) as the source of a mechanism-design result, independent of the specific banking application.

Questions to Test Understanding

Comprehension — Can you state, without notation, why a demand deposit contract with $r_1 = 1$ is never subject to a run, and why it also provides no liquidity insurance? **Analysis —** Can you explain, in your own words, exactly which assumption Proposition 1's proof uses to rule out a constant $T = 1$ payment across realizations of t, and why that same argument does not apply to the government's tax scheme in Proposition 2? **Research —** Can you design an empirical test, using data the paper itself does not use, that would distinguish a Diamond-Dybvig coordination-failure run from a Fisher-style insolvency-driven run in a specific historical or contemporary episode?

Papers to Read Next

Diamond (1984), "Financial Intermediation and Delegated Monitoring," for the complementary lending-side theory of banking; Jacklin and Bhattacharya (1988) for the panics-versus-information distinction that operationalizes §8.3 above; Goldstein and Pauzner (2005) for the equilibrium-selection refinement that resolves the single most important weakness identified in §7.3.

13. Conclusion

ONE-SENTENCE THESIS

Demand deposits solve a genuine, privately-informed liquidity-insurance problem that no competitive market can solve directly, but the very contractual feature that delivers this benefit — first-come-first-served payment — simultaneously and unavoidably creates a second, self-fulfilling equilibrium in which the bank fails regardless of its underlying soundness.

THREE MOST IMPORTANT INSIGHTS

First, liquidity demand can be derived from privately observed preference heterogeneity rather than assumed, giving banking theory genuine micro-foundations. Second, the coexistence of an efficient equilibrium and a run equilibrium is a structural property of any contract subject to sequential service, not a symptom of mismanagement, meaning fragility and liquidity provision are two faces of the same institutional choice. Third, what allows government deposit insurance to dominate any privately feasible arrangement is not superior resources but a different, and structurally unavailable-to-banks, information and enforcement technology — the ability to condition payment on the full ex post realization of aggregate withdrawals rather than only on an individual's place in line.

STRONGEST EVIDENCE

Since this is a theoretical paper, its strongest "evidence" is internal logical validity: Proposition 1's impossibility result is a clean, general proof covering the entire class of sequential-service contracts, not merely the specific demand-deposit form, which is what elevates it from an example to a theorem.

BIGGEST WEAKNESS

The absence of any theory of equilibrium selection: the model shows a bad equilibrium can exist but is silent on when or why an economy actually coordinates on it, which is the central predictive gap the subsequent global-games literature was built to close.

MOST IMPORTANT RESEARCH GAP

A general, real-time (not merely after-the-fact) method for distinguishing coordination-failure runs from fundamentals-driven runs remains unresolved, and matters directly for whether deposit insurance and suspension — the paper's own prescriptions — are the right tool for a given episode.

BEST RESEARCH OPPORTUNITY

Testing the uninsured-deposit comparative static (Research Idea 2, §11) against the 2023 U.S. regional-bank episode, because it directly confronts the paper's own central normative claim with a clean, well-measured, recently available natural experiment.

THREE PAPERS TO READ NEXT

Diamond (1984); Goldstein and Pauzner (2005); Jacklin and Bhattacharya (1988).

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